

Implicit Differentiation and Inverse Functions

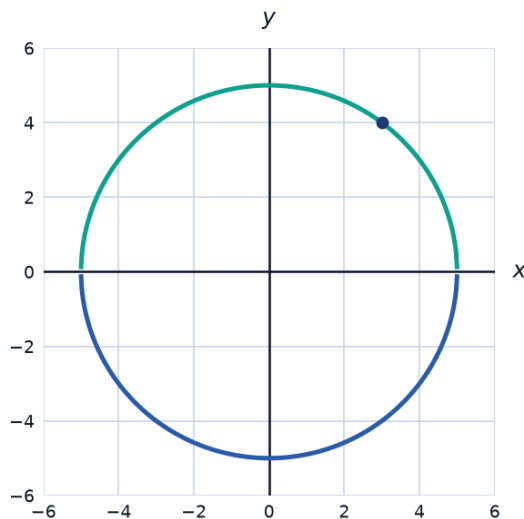


Implicit differentiation basics

- 1 Find $\frac{dy}{dx}$ for $x^2 + y^2 = 25$.
- 2 Find $\frac{dy}{dx}$ for $x^3 + y^3 = 6xy$.
- 3 Find $\frac{dy}{dx}$ for $\sin(xy) = x$.

Tangent lines to implicit curves

- 4 Find the equation of the tangent line to $x^2 + y^2 = 25$ at the point $(3, 4)$.



- 5 Find $\frac{dy}{dx}$ at the point $(1, 2)$ on the curve $x^2y + y^3 = 10$.

Second derivatives via implicit differentiation

- 6 Find $\frac{d^2y}{dx^2}$ for $x^2 + y^2 = 1$, expressing your answer in terms of x and y .

Derivatives of inverse functions

- 7 If $f(x) = x^3 + 2x + 1$, find $(f^{-1})'(4)$, using $(f^{-1})'(b) = \frac{1}{f'(f^{-1}(b))}$ and noting $f(1) = 4$.
- 8 State the derivatives of the inverse trigonometric functions $\sin^{-1} x$, $\cos^{-1} x$, and $\tan^{-1} x$.

- 9 Differentiate $f(x) = \tan^{-1}(3x)$.
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Implicit differentiation with logarithmic and exponential terms

- 10 Find $\frac{dy}{dx}$ for $e^y = xy$.
- 11 Find $\frac{dy}{dx}$ for $\ln(x + y) = x^2$.
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Related applications of implicit differentiation

- 12 Find all points on the curve $x^2 + xy + y^2 = 7$ where the tangent line is horizontal.
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Finding vertical tangents implicitly

- 13 Find the points on the circle $x^2 + y^2 = 9$ where the tangent line is vertical (i.e. where $\frac{dx}{dy} = 0$, or equivalently where $\frac{dy}{dx}$ is undefined).
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Inverse function derivatives from a graph or table

- 14 A function f is one-to-one and differentiable, with $f(3) = 10$ and $f'(3) = 4$. Find $(f^{-1})'(10)$.
- 15 If $g(x) = x^5 + 2x - 3$, find $(g^{-1})'(-3)$, first identifying the value a such that $g(a) = -3$.
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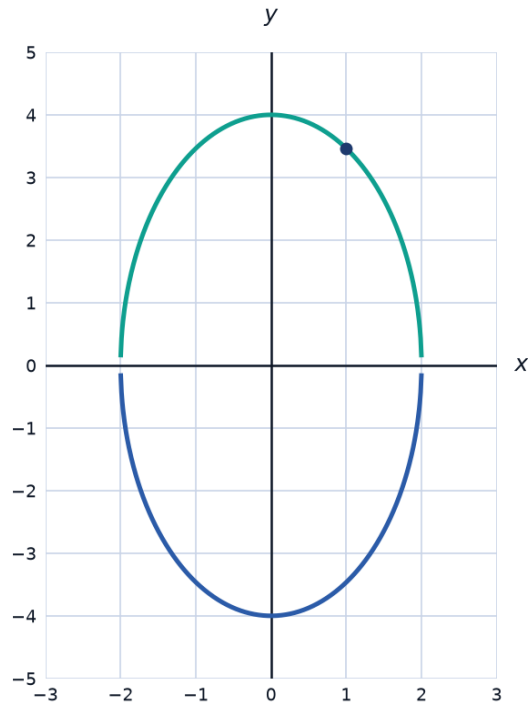
Implicit differentiation in an applied context

- 16 A conical tank has height and radius related by $h = 2r$ at all times as it fills, so its volume is $V = \frac{1}{3}\pi r^2(2r) = \frac{2}{3}\pi r^3$. Using implicit differentiation with respect to time, find $\frac{dV}{dt}$ in terms of r and $\frac{dr}{dt}$.
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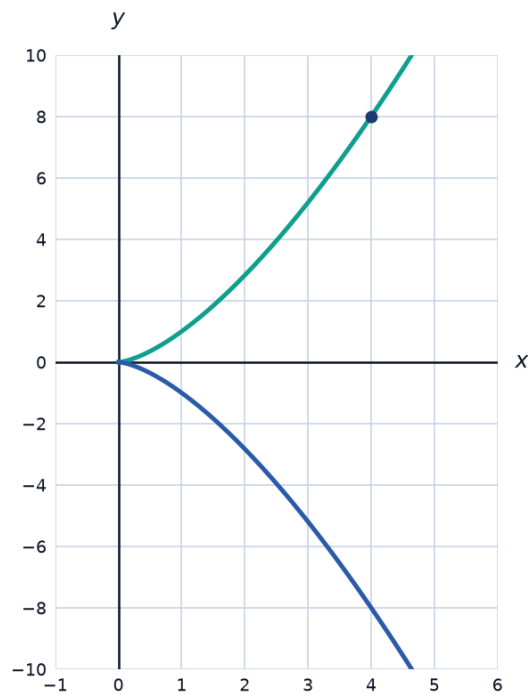
A capstone implicit differentiation problem

- 17 Find $\frac{dy}{dx}$ for the curve $x^2y^2 + xy = 6$ at the point $(1, 2)$.
- 18 Find $\frac{dy}{dx}$ for $\cos(x - y) = y\sin x$.

- 19 A satellite dish's cross-section satisfies $4x^2 + y^2 = 16$ (an ellipse). Find $\frac{dy}{dx}$ at the point $(1, 2\sqrt{3})$ on the curve.



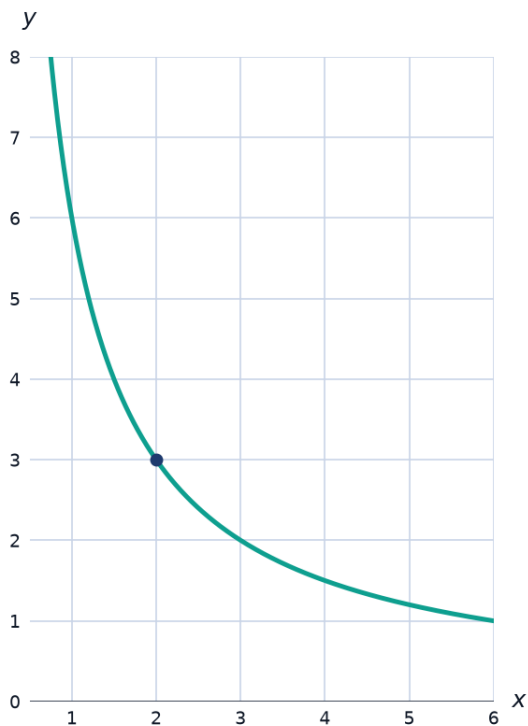
- 20 For the curve $y^2 = x^3$ (a cuspidal cubic), complete each step:



- Find $\frac{dy}{dx}$ implicitly.
- Evaluate $\frac{dy}{dx}$ at the point $(4, 8)$.
- Find the equation of the tangent line at that point.

21 If $f(x) = 2x^3 + x - 5$ is one-to-one and differentiable, find $(f^{-1})'(-2)$, first identifying the input a such that $f(a) = -2$.

22 For the hyperbola $xy = 6$, complete each step:



- a Find $\frac{dy}{dx}$ using implicit differentiation.
- b Confirm your answer matches what you'd get by first solving for $y = \frac{6}{x}$ and differentiating directly.
- c Evaluate $\frac{dy}{dx}$ at the point (2, 3).

Multiple choice — easy 15

23 Find $\frac{dy}{dx}$ for $x^2 + 5y^2 = 126$ at the point $(-1, 5)$.

- a. $\frac{1}{25}$
- b. $\frac{1}{5}$
- c. 1
- d. $-\frac{1}{25}$

24 Find $\frac{dy}{dx}$ for $x^2 + 5y^2 = 30$ at the point $(-5, 1)$.

a. 5

b. -1

c. 25

d. 1

25 Find $\frac{dy}{dx}$ for $x^2 + 4y^2 = 20$ at the point $(2, 2)$.

a. -1

b. $\frac{1}{4}$

c. -4

d. $-\frac{1}{4}$

26 Find $\frac{dy}{dx}$ for $x^2 + y^2 = 41$ at the point $(4, 5)$.

a. $\frac{4}{5}$

b. $\frac{1}{5}$

c. $-\frac{3}{10}$

d. $-\frac{4}{5}$

27 Find $\frac{dy}{dx}$ for $x^2 + y^2 = 20$ at the point $(2, 4)$.

a. $-\frac{1}{2}$

b. $\frac{1}{2}$

c. -1

d. 0

28 Find $\frac{dy}{dx}$ for $x^2 + 5y^2 = 141$ at the point $(4, -5)$.

a. $-\frac{4}{25}$

b. $\frac{4}{5}$

c. $\frac{4}{25}$

d. 4

29 Find $\frac{dy}{dx}$ for $x^2 + y^2 = 29$ at the point $(-5, 2)$.

a. 3

b. $\frac{5}{2}$

c. $-\frac{5}{2}$

d. 2

30 Find $\frac{dy}{dx}$ for $x^2 + y^2 = 41$ at the point $(-5, 4)$.

a. $\frac{9}{4}$

b. $\frac{7}{4}$

c. $-\frac{5}{4}$

d. $\frac{5}{4}$

31 Find $\frac{dy}{dx}$ for $x^2 + y^2 = 13$ at the point $(-3, 2)$.

a. $-\frac{3}{2}$

b. 2

c. $\frac{3}{2}$

d. 1

32 Find $\frac{dy}{dx}$ for $x^2 + 4y^2 = 40$ at the point $(-2, -3)$.

a. $\frac{1}{6}$

b. $-\frac{2}{3}$

c. $-\frac{8}{3}$

d. $-\frac{1}{6}$

33 Find $\frac{dy}{dx}$ for $x^2 + 4y^2 = 8$ at the point $(2, 1)$.

a. $-\frac{1}{2}$

b. -2

c. -8

d. $\frac{1}{2}$

34 Find $\frac{dy}{dx}$ for $x^2 + 3y^2 = 28$ at the point $(1, 3)$.

a. $\frac{1}{9}$

b. -1

c. $-\frac{1}{9}$

d. $-\frac{1}{3}$

35 Find $\frac{dy}{dx}$ for $x^2 + 3y^2 = 64$ at the point $(4, -4)$.

a. $-\frac{1}{3}$

b. 3

c. $\frac{1}{3}$

d. 1

- 36 Find $\frac{dy}{dx}$ for $x^2 + 3y^2 = 28$ at the point $(5, 1)$.
- a. -5
 - b. -15
 - c. $\frac{5}{3}$
 - d. $-\frac{5}{3}$
- 37 Find $\frac{dy}{dx}$ for $x^2 + 4y^2 = 89$ at the point $(-5, -4)$.
- a. $\frac{5}{16}$
 - b. -5
 - c. $-\frac{5}{4}$
 - d. $-\frac{5}{16}$
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Multiple choice — medium 15

- 38 Find $\frac{dy}{dx}$ for $xy^2 + x^2 = 12$ at the point $(2, -2)$.
- a. -1
 - b. 2
 - c. 1
 - d. 0
- 39 Find $\frac{dy}{dx}$ for $xy^2 + x^2 = 70$ at the point $(5, -3)$.
- a. $\frac{1}{30}$
 - b. $\frac{19}{15}$
 - c. $\frac{19}{30}$
 - d. $-\frac{19}{30}$

40 Find $\frac{dy}{dx}$ for $xy^2 + x^2 = 26$ at the point $(1, -5)$.

a. $-\frac{27}{10}$

b. $-\frac{23}{10}$

c. $\frac{27}{10}$

d. $\frac{27}{5}$

41 Find $\frac{dy}{dx}$ for $xy^2 + x^2 = 12$ at the point $(-4, 1)$.

a. $\frac{7}{8}$

b. $-\frac{7}{4}$

c. $-\frac{9}{8}$

d. $-\frac{7}{8}$

42 Find $\frac{dy}{dx}$ for $xy^2 + x^2 = 17$ at the point $(1, 4)$.

a. $\frac{9}{4}$

b. $-\frac{9}{4}$

c. $\frac{7}{4}$

d. $-\frac{9}{2}$

43 Find $\frac{dy}{dx}$ for $xy^2 + x^2 = 5$ at the point $(-5, 2)$.

a. $-\frac{3}{5}$

b. $-\frac{3}{10}$

c. $-\frac{7}{10}$

d. $\frac{3}{10}$

44 Find $\frac{dy}{dx}$ for $xy^2 + x^2 = -39$ at the point $(-3, 4)$.

a. $\frac{5}{6}$

b. $-\frac{11}{12}$

c. $\frac{5}{12}$

d. $-\frac{5}{12}$

45 Find $\frac{dy}{dx}$ for $xy^2 + x^2 = 12$ at the point $(-4, -1)$.

a. $\frac{7}{4}$

b. $\frac{7}{8}$

c. $\frac{9}{8}$

d. $-\frac{7}{8}$

46 Find $\frac{dy}{dx}$ for $xy^2 + x^2 = 21$ at the point $(3, 2)$.

a. $-\frac{1}{6}$

b. $-\frac{5}{3}$

c. $\frac{5}{6}$

d. $-\frac{5}{6}$

47 Find $\frac{dy}{dx}$ for $xy^2 + x^2 = 45$ at the point $(5, 2)$.

a. $-\frac{7}{10}$

b. $\frac{7}{10}$

c. $-\frac{7}{5}$

d. $-\frac{3}{10}$

48 Find $\frac{dy}{dx}$ for $xy^2 + x^2 = -55$ at the point $(-5, -4)$.

a. $-\frac{3}{10}$

b. $\frac{3}{20}$

c. $\frac{13}{20}$

d. $-\frac{3}{20}$

49 Find $\frac{dy}{dx}$ for $xy^2 + x^2 = 54$ at the point $(2, 5)$.

a. $\frac{29}{20}$

b. $\frac{21}{20}$

c. $-\frac{29}{10}$

d. $-\frac{29}{20}$

50 Find $\frac{dy}{dx}$ for $xy^2 + x^2 = 6$ at the point $(-3, -1)$.

a. $\frac{7}{6}$

b. $-\frac{5}{6}$

c. $\frac{5}{6}$

d. $\frac{5}{3}$

51 Find $\frac{dy}{dx}$ for $xy^2 + x^2 = 54$ at the point $(2, -5)$.

a. $\frac{29}{20}$

b. $-\frac{21}{20}$

c. $\frac{29}{10}$

d. $-\frac{29}{20}$

52 Find $\frac{dy}{dx}$ for $xy^2 + x^2 = 0$ at the point $(-4, 2)$.

a. $-\frac{1}{2}$

b. $-\frac{3}{4}$

c. $\frac{1}{4}$

d. $-\frac{1}{4}$

Multiple choice — hard 15

53 If $f(x) = 2x^3 + 6x + 5$ is one-to-one and differentiable, find $(f^{-1})'(5)$, using $f(0) = 5$.

a. $\frac{1}{6}$

b. $\frac{1}{7}$

c. $-\frac{1}{6}$

d. 6

54 If $f(x) = 3x^3 - 4x + 3$ is one-to-one and differentiable, find $(f^{-1})'(-66)$, using $f(-3) = -66$.

a. $-\frac{1}{77}$

b. 77

c. $\frac{1}{77}$

d. $\frac{1}{78}$

55 If $f(x) = 2x^3 + 6x - 3$ is one-to-one and differentiable, find $(f^{-1})'(69)$, using $f(3) = 69$.

a. 60

b. $-\frac{1}{60}$

c. $\frac{1}{61}$

d. $\frac{1}{60}$

56 If $f(x) = x^3 + x + 3$ is one-to-one and differentiable, find $(f^{-1})'(13)$, using $f(2) = 13$.

a. 13

b. $-\frac{1}{13}$

c. $\frac{1}{13}$

d. $\frac{1}{14}$

57 If $f(x) = 3x^3 - 4x + 2$ is one-to-one and differentiable, find $(f^{-1})'(2)$, using $f(0) = 2$.

a. $-\frac{1}{4}$

b. $\frac{1}{4}$

c. $-\frac{1}{3}$

d. -4

58 Find $\frac{dy}{dx}$ for $x^2y + y^3 = -68$ at the point $(1, -4)$.

a. $-\frac{33}{98}$

b. $-\frac{8}{49}$

c. $\frac{8}{49}$

d. $-\frac{8}{47}$

59 Find $\frac{dy}{dx}$ for $x^2y + y^3 = -10$ at the point $(-3, -1)$.

a. 0

b. $-\frac{1}{2}$

c. -1

d. $\frac{1}{2}$

60 Find $\frac{dy}{dx}$ for $x^2y + y^3 = -16$ at the point $(-2, -2)$.

a. $\frac{1}{2}$

b. -1

c. 1

d. $-\frac{1}{2}$

61 Find $\frac{dy}{dx}$ for $x^2y + y^3 = 17$ at the point $(-4, 1)$.

a. $\frac{8}{19}$

b. $-\frac{8}{19}$

c. $\frac{8}{13}$

d. $\frac{27}{19}$

62 Find $\frac{dy}{dx}$ for $x^2y + y^3 = 10$ at the point $(-1, 2)$.

a. $-\frac{4}{13}$

b. $-\frac{4}{11}$

c. $\frac{4}{13}$

d. $-\frac{5}{26}$

63 Find $\frac{dy}{dx}$ for $x^2 + xy + y^2 = 19$ at the point $(-2, -3)$.

a. $\frac{7}{8}$

b. $-\frac{7}{5}$

c. $-\frac{7}{8}$

d. $\frac{1}{8}$

64 Find $\frac{dy}{dx}$ for $x^2 + xy + y^2 = 1$ at the point $(1, -1)$.

a. -3

b. -1

c. 1

d. $\frac{1}{2}$

65 Find $\frac{dy}{dx}$ for $x^2 + xy + y^2 = 7$ at the point $(3, -2)$.

a. -8

b. $\frac{9}{2}$

c. 4

d. -4

66 Find $\frac{dy}{dx}$ for $x^2 + xy + y^2 = 7$ at the point $(-1, -2)$.

a. 0

b. $-\frac{4}{5}$

c. $-\frac{4}{3}$

d. $\frac{4}{5}$

67 Find $\frac{dy}{dx}$ for $x^2 + xy + y^2 = 7$ at the point $(1, 2)$.

a. $-\frac{4}{3}$

b. $-\frac{4}{5}$

c. 0

d. $\frac{4}{5}$